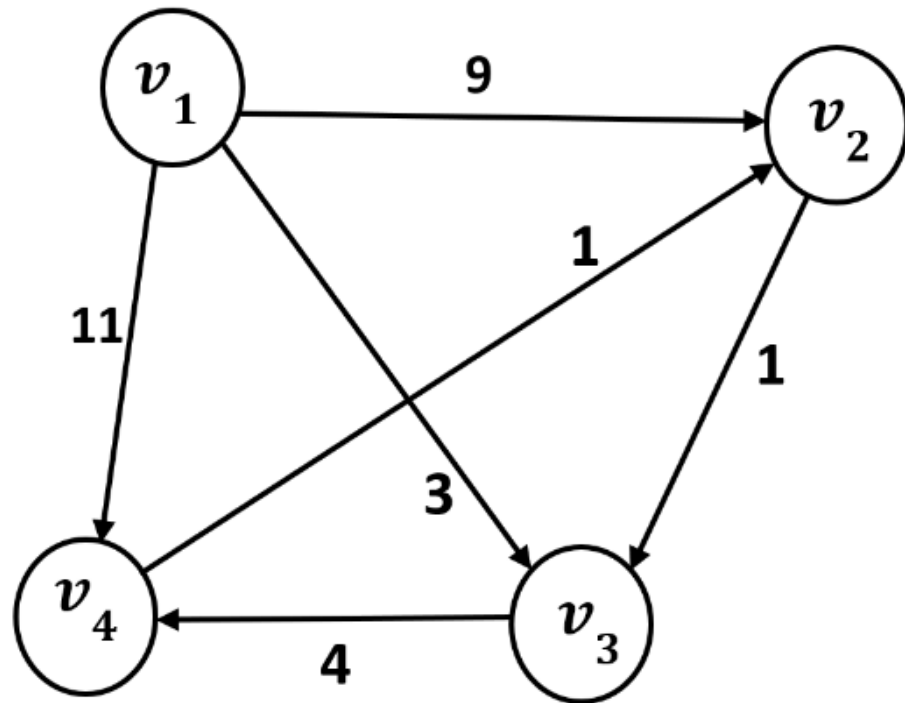


Greedy+DP

Find the all shortest paths from v_1 to the other vertices using Dijkstra Algorithm:



```

Dijkstra(W)
  FOR i=2 TO n DO
    D[i]=W[1][i]
    P[i]=1
  FOR i=1 TO n-1 DO
    m=index of minimum in D
    FOR EACH j in D:
      IF D[m]+W[m][j]<D[j] THEN
        D[j]=D[m]+W[m][j]
        P[j]=m
    PRINT D[m]
    D.delete(m)

```

$$D^{new}[i] = \min\{D^{old}, [i] \ D^{old}[m] + W[m][i]\}$$

	V2	V3	V4
D	9	3	11

	V2	V3	V4
P	1	1	1

m=3

$$D[2] = \min\{D[2], D[3] + W[3][2]\} = \min\{9, 3 + \text{inf}\} = 9$$

$$D[4] = \min\{D[4], D[3] + W[3][4]\} = \min\{11, 3 + 4\} = 7$$

	V2	V3	V4
D	9	3	7

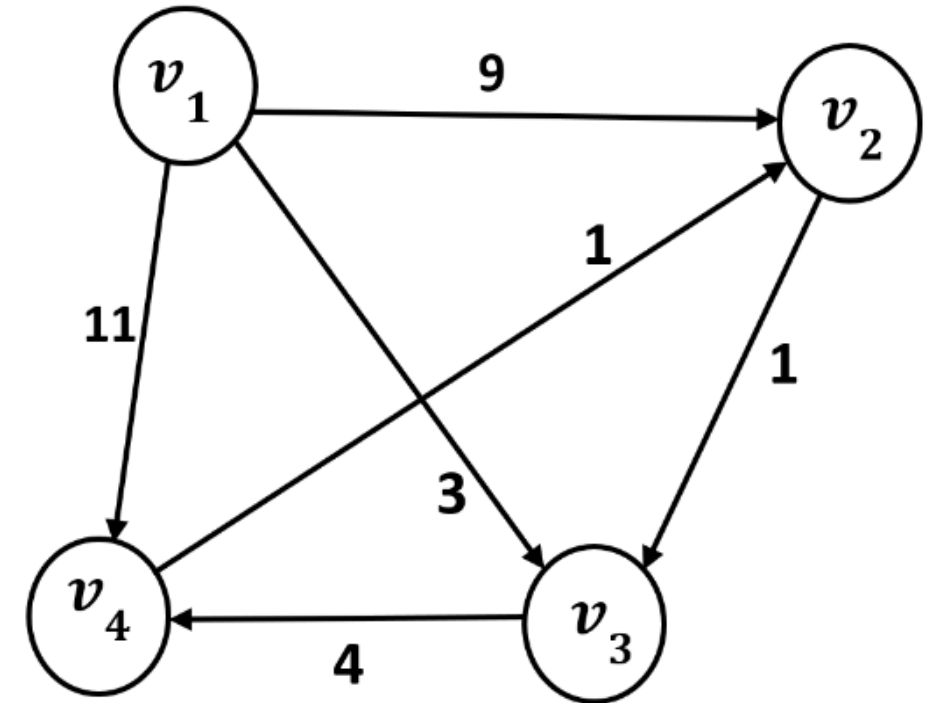
	V2	V3	V4
P	1	1	3

m=4

$$D[2] = \min\{D[2], D[4] + W[4][2]\} = \min\{9, 7 + 1\} = 8$$

	V2	V3	V4
D	8	3	7

	V2	V3	V4
P	4	1	3



Calculating shortest paths using P matrix:

	V2	V3	V4
P	4	1	3

- $SP(1,2)=(1,\dots,2)=(1,\dots,4,2)=(1,3,4,2)$
- $SP(1,3)=(1,3)$
- $SP(1,4)=(1,\dots,4)=1,3,4$

Binomial Coefficients/ Pascal triangle

n=0				1						
n=1				1			1			
n=2			1		2		1			
n=3		1		3		3		1		
n=4		1		4		6		4		1
n=5	1		5		10		10		5	

$$(x + y)^n = \sum_{r=0}^n \binom{n}{r} x^r y^{n-r}$$

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$$

$$0 \leq r \leq n$$

$$\binom{n}{0} = 1$$

$$\binom{r}{r} = 1$$

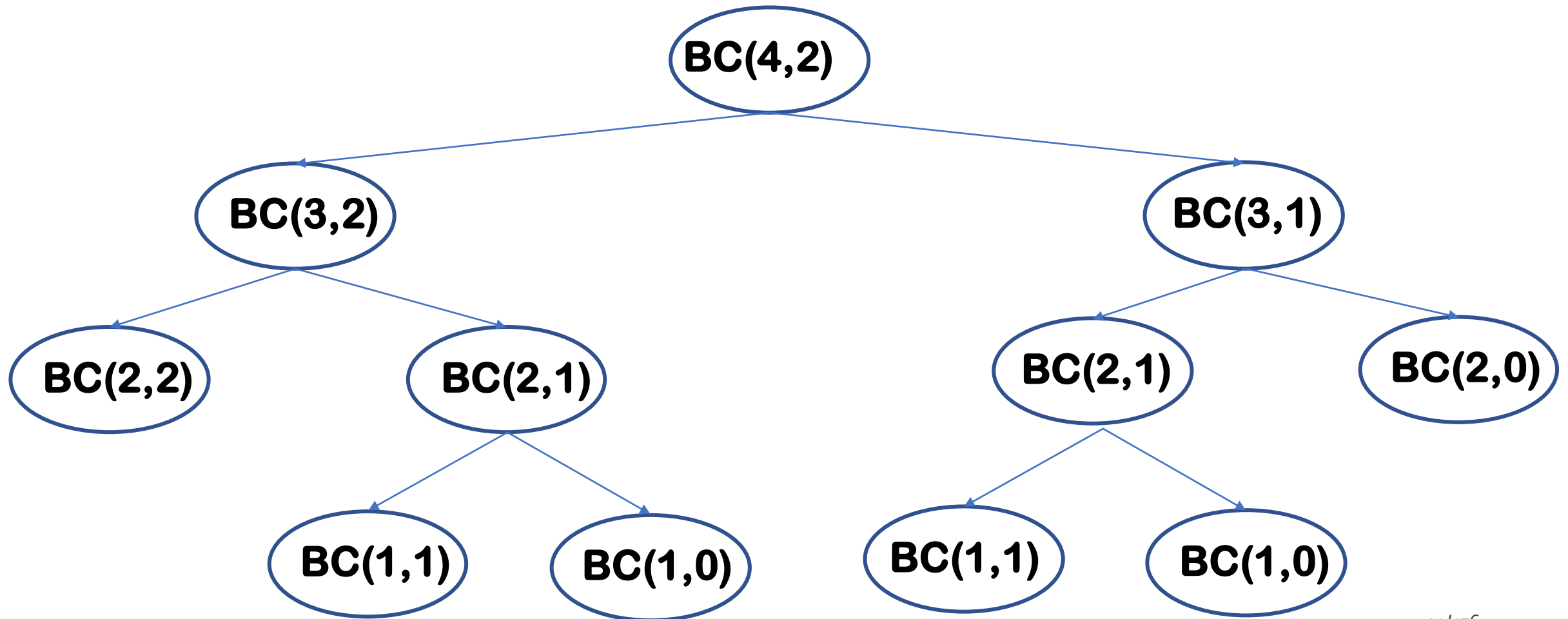
Binomial Coefficients: without DP

- $n \geq r$: $BC(r, r) = BC(n, 0) = 1$, $BC(n, r) = BC(n - 1, r - 1) + BC(n - 1, r)$

```
BC(n, r):  
    if r==0 or n==r then  
        return 1  
    return BC(n-1, r)+BC(n-1, r-1)
```

Exponential complexity!

Binomial Coefficient



Binomial Coefficient: using DP

$$M[i][j] = \binom{i}{j}, j \leq i$$

	0	1	...	r
0				
1				
2				
:				
n				

Binomial Coefficient

$$\binom{i}{j} = \binom{i-1}{j-1} + \binom{i-1}{j}, \quad \binom{i}{0} = 1, \quad \binom{j}{j} = 1$$

$$M[i][j] = M[i-1][j-1] + M[i-1][j],$$

$$M[i][0] = 1, \quad M[j][j] = 1$$

$$M[i][j] = \binom{i}{j}, j \leq i$$

	0	1	...	r
0	1			
1	1	1		
2	1		1	
:	1			
n	1			

Binomial Coefficient

```
BC(n,r):  
  for i=0 to n do  
    for j=0 to r do  
      if i>=j then  
        if j==0 or i==j then  
          M[i][j]=1  
        else  
          M[i][j]=M[i-1][j]+M[i-1][j-1]  
  return M[n][r]
```

$O(nr)$

$$M[i][j] = \binom{i}{j}, j \leq i$$

	0	1	2
0	$\binom{0}{0}$		
1	$\binom{1}{0}$	$\binom{1}{1}$	
2	$\binom{2}{0}$	$\binom{2}{1}$	$\binom{2}{2}$
3	$\binom{3}{0}$	$\binom{3}{1}$	$\binom{3}{2}$
4	$\binom{4}{0}$	$\binom{4}{1}$	$\binom{4}{2}$

Floyd-Warshall

- You are given a weighted directed graph G with n nodes labeled from 1 to n . We apply the Floyd algorithm to calculate the length of the shortest path between all pairs of nodes. Let P be the resulting path matrix. Write pseudocode that generates the shortest path between all pairs of nodes using P . Note that for $1 \leq i, j \leq n, P[i][j] = k$ means that the shortest path from i to j passes through k . if $P[i][j] = 0$, edge (i,j) is the shortest path from i to j .

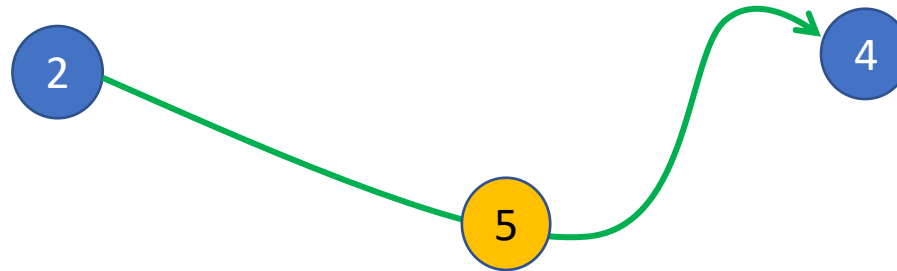
Floyd-Warshall

	1	2	3	4	5
1	0	0	4	0	3
2	4	0	0	5	3
3	0	1	0	0	0
4	0	1	2	0	0
5	0	0	4	1	0



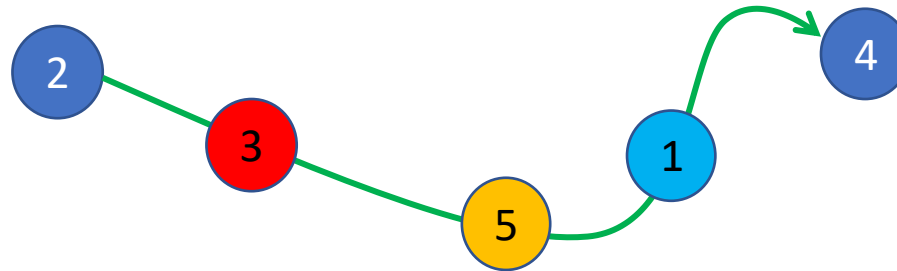
Floyd-Warshall

	1	2	3	4	5
1	0	0	4	0	3
2	4	0	0	5	3
3	0	1	0	0	0
4	0	1	2	0	0
5	0	0	4	1	0



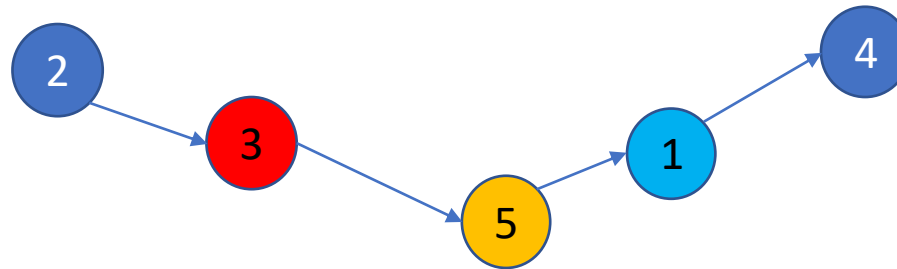
Floyd-Warshall

	1	2	3	4	5
1	0	0	4	0	3
2	4	0	0	5	3
3	0	1	0	0	0
4	0	1	2	0	0
5	0	0	4	1	0



Floyd-Warshall

	1	2	3	4	5
1	0	0	4	0	3
2	4	0	0	5	3
3	0	1	0	0	0
4	0	1	2	0	0
5	0	0	4	1	0



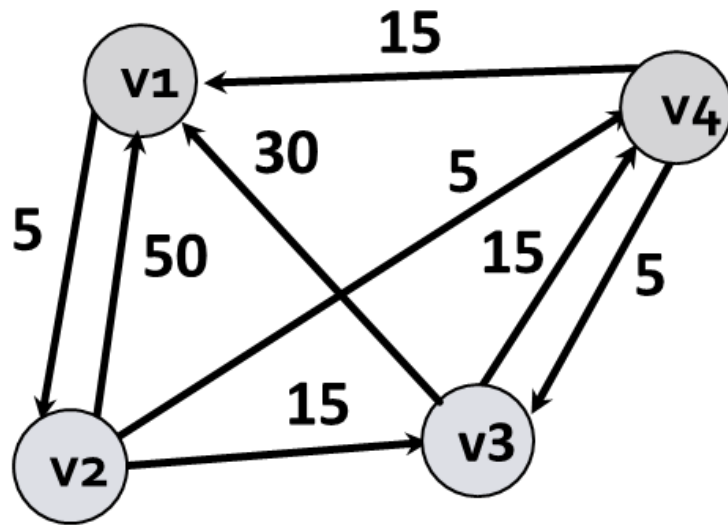
Floyd-Warshall

```
shortest_path(i, j, P):  
    path = ""  
    k = P[i][j]  
    if k == 0 then  
        return ""  
    path += shortest_path(i, k, P)  
    path += str(k)  
    path += shortest_path(k, j, P)  
    return path
```

```
for i=1 to n do  
    for j=1 to n do  
        path=""  
        path += str(i)  
        path += shortest_path(i, j, P)  
        path += str(j)  
        print(path)
```

Floyd-Warshall

- Find all shortest paths between all pairs by using Floyd algorithm.



$$W = \begin{bmatrix} 0 & 5 & \infty & \infty \\ 50 & 0 & 15 & 5 \\ 30 & \infty & 0 & 15 \\ 15 & \infty & 5 & 0 \end{bmatrix}$$

Floyd-Warshall algorithm

```
FloydWarshall(W[][],n)
  D=W
  P=0 //initializing matrix P
  for k=1 to n do
    for i=1 to n do
      for j=1 to n do
        if D[i][k]+D[k][j]<D[i][j] then
          D[i][j]=D[i][k]+D[k][j]
          P[i][j]=k
  return D
```

D_0	v1	v2	v3	v4
v1	0	5	∞	∞
v2	50	0	15	5
v3	30	∞	0	15
v4	15	∞	5	0

P	v1	v2	v3	v4
v1	0	0	0	0
v2	0	0	0	0
v3	0	0	0	0
v4	0	0	0	0

$$D^{(k+1)}[i][j] = \min\{D^{(k)}[i][k+1] + D^{(k)}[k+1][j], D^{(k)}[i][j]\}$$

NOTE: In the following calculations:

- 1) The red highlights indicate where the shortest path is updated. These will be used to construct the path matrix.
- 2) The purple highlights indicate values that do not need to be computed, as they will not change.

K=1

1. $D[1][j] = \min\{D[1][j], D[1][1] + D[1][j]\} = D[1][j]$ for $j=2,3,4$
2. $D[2][j] = \min\{D[2][j], D[2][1] + D[1][j]\} = \min\{D[2][j], 50 + D[1][j]\}$, for $j=1,3,4$
 $D[2][1]=50$, $D[2][3]=15$, $D[2][4]=5$
3. $D[3][j] = \min\{D[3][j], D[3][1] + D[1][j]\} = \min\{D[3][j], 30 + D[1][j]\}$, for $j=1,2,4$
 $D[3][1]=30$, $D[3][2]=35$, $D[3][4]=15$
4. $D[4][j] = \min\{D[4][j], D[4][1] + D[1][j]\} = \min\{D[4][j], 15 + D[1][j]\}$, for $j=1,2,3$
 $D[4][1]=15$, $D[4][2]=20$, $D[4][3]=5$

D_1	v1	v2	v3	v4
v1	0	5	∞	∞
v2	50	0	15	5
v3	30	35	0	15
v4	15	20	5	0

P	v1	v2	v3	v4
v1	0	0	0	0
v2	0	0	0	0
v3	0	1	0	0
v4	0	1	0	0

K=2

1. $D[1][j] = \min\{D[1][j], D[1][2] + D[2][j]\} = \min\{D[1][j], 5 + D[2][j]\}$ for $j=2,3,4$
 $D[1][2]=5$, $D[1][3]=20$, $D[1][4]=10$
2. $D[2][j] = \min\{D[2][j], D[2][2] + D[2][j]\} = D[2][j]$ for $j=1,3,4$
3. $D[3][j] = \min\{D[3][j], D[3][2] + D[2][j]\} = \min\{D[3][j], 35 + D[2][j]\}$ for $j=1,2,4$
 $D[3][1]=30$, $D[3][2]=35$, $D[3][4]=15$
4. $D[4][j] = \min\{D[4][j], D[4][2] + D[2][j]\} = \min\{D[4][j], 20 + D[2][j]\}$, for $j=1,2,3$
 $D[4][1]=15$, $D[4][2]=20$, $D[4][3]=5$

D_2	v1	v2	v3	v4
v1	0	5	20	10
v2	50	0	15	5
v3	30	35	0	15
v4	15	20	5	0

P	v1	v2	v3	v4
v1	0	0	2	2
v2	0	0	0	0
v3	0	1	0	0
v4	0	1	0	0

K=3

1. $D[1][j] = \min\{D[1][j], D[1][3] + D[3][j]\} = \min\{D[1][j], 20 + D[3][j]\}$ for $j=2,3,4$
 $D[1][2]=5, D[1][3]=20, D[1][4]=10$
2. $D[2][j] = \min\{D[2][j], D[2][3] + D[3][j]\} = \min\{D[2][j], 15 + D[3][j]\}$ for $j=1,3,4$
 $D[2][1]=45, D[2][3]=15, D[2][4]=5$
3. $D[3][j] = \min\{D[3][j], D[3][3] + D[3][j]\} = D[3][j]$ for $j=1,2,4$
4. $D[4][j] = \min\{D[4][j], D[4][3] + D[3][j]\} = \min\{D[4][j], 5 + D[3][j]\}$, for $j=1,2,3$
 $D[4][1]=15, D[4][2]=20, D[4][3]=5$

D_3	v1	v2	v3	v4
v1	0	5	20	10
v2	45	0	15	5
v3	30	35	0	15
v4	15	20	5	0

P	v1	v2	v3	v4
v1	0	0	2	2
v2	3	0	0	0
v3	0	1	0	0
v4	0	1	0	0

K=4

1. $D[1][j] = \min\{D[1][j], D[1][4] + D[4][j]\} = \min\{D[1][j], 10 + D[4][j]\}$ for $j=2,3,4$
 $D[1][2]=5$, **$D[1][3]=15$** , $D[1][4]=10$
2. $D[2][j] = \min\{D[2][j], D[2][4] + D[4][j]\} = \min\{D[2][j], 5 + D[4][j]\}$, for $j=1,3,4$
 $D[2][1]=20$, **$D[2][3]=10$** , $D[2][4]=5$
3. $D[3][j] = \min\{D[3][j], D[3][4] + D[4][j]\} = \min\{D[3][j], 15 + D[4][j]\}$ for $j=1,2,4$
 $D[3][1]=30$, $D[3][2]=35$, $D[3][4]=15$
4. $D[4][j] = \min\{D[4][j], D[4][4] + D[4][j]\} = D[4][j]$ for $j=1,2,3$

D_4	v1	v2	v3	v4
v1	0	5	15	10
v2	20	0	10	5
v3	30	35	0	15
v4	15	20	5	0

P	v1	v2	v3	v4
v1	0	0	4	2
v2	4	0	4	0
v3	0	1	0	0
v4	0	1	0	0

Calculating shortest paths using P matrix

- v1 to v2: Direct path, $v1 \rightarrow v2$
- v1 to v3: Passes through v4 : $v1 \rightarrow v2 \rightarrow v4 \rightarrow v3$
- v1 to v4: Passes through v2 : $v1 \rightarrow v2 \rightarrow v4$
- v2 to v1: Passes through v4, $v2 \rightarrow v4 \rightarrow v1$
- v2 to v3: Passes through v4, $v2 \rightarrow v4 \rightarrow v3$
- v2 to v4: Direct path, $v2 \rightarrow v4$
- v3 to v1: Direct path, $v3 \rightarrow v1$
- v3 to v2: Passes through v1, $v3 \rightarrow v1 \rightarrow v2$
- v3 to v4: Direct path, $v3 \rightarrow v4$
- v4 to v1: Direct path, $v4 \rightarrow v1$
- v4 to v2: Passes through v1, $v4 \rightarrow v1 \rightarrow v2$
- v4 to v3: Direct path, $v4 \rightarrow v3$

P	v1	v2	v3	v4
v1	0	0	4	2
v2	4	0	4	0
v3	0	1	0	0
v4	0	1	0	0